H

H
H

H
H

H

H

H
H

H
H

H
H

H
H

r l ar fft

H

H
H

A fft Aft c l

H

H
H

H

H

H

H

A fft fffft

H
H

l fft fftr fft

R fft

l r c fatc

H

H

		H

H

H

H

H

l r c ffatc c fftc Al l c r

-
- H
-
-
-

c co ffffftc

-
- H
-
-

H
H

fft fft c fatc fft l R

Name	Date modified	Type	Size
a32_startup	7/11/2023 4:15 PM	File folder	
build	7/7/2023 7:51 AM	File folder	
drivers	7/7/2023 7:51 AM	File folder	
example	7/11/2023 4:15 PM	File folder	
include	7/11/2023 6:01 AM	File folder	
lib	7/7/2023 7:51 AM	File folder	
RTE	7/11/2023 4:15 PM	File folder	
services_lib	7/11/2023 4:15 PM	File folder	
A32_MRAM	7/11/2023 4:15 PM	Windows Script C...	
A32_TCM	7/11/2023 4:15 PM	Windows Script C...	
CMakeLists	7/11/2023 4:15 PM	Text Document	
CMakePresets	7/11/2023 4:15 PM	JSON File	
gcc_A32_MRAM	7/11/2023 4:15 PM	LD File	
gcc_A32_TCM	7/11/2023 4:15 PM	LD File	
gcc_M55_HE_MRAM	7/11/2023 4:15 PM	LD File	
gcc_M55_HE_TCM	7/11/2023 4:15 PM	LD File	
gcc_M55_HP_MRAM	7/11/2023 4:15 PM	LD File	
gcc_M55_HP_TCM	7/11/2023 4:15 PM	LD File	
License	7/7/2023 7:51 AM	Text Document	
M55_HE_MRAM	7/11/2023 4:15 PM	Windows Script C...	
M55_HE_TCM	7/11/2023 4:15 PM	Windows Script C...	
M55_HP_MRAM	7/11/2023 4:15 PM	Windows Script C...	
M55_HP_TCM	7/11/2023 4:15 PM	Windows Script C...	
Makefile_linux	7/7/2023 7:51 AM	File	
README	7/11/2023 4:15 PM	Markdown Source...	
toolchain-armclang	7/11/2023 4:15 PM	CMake Source File	
toolchain-gnu	7/11/2023 4:15 PM	CMake Source File	

H
H

fft fft ffrfft R A

fft fft ffrfft A

A fft c c c afcft

-
-

A a c

H
H

H

H Al dfft rfflft

-
-
-

-
-
-
-
-

	✓	
		✓
	✓	
	✓	
	✓	
		✓
	✓	
		✓
	✓	
		✓
	✓	✓

l c c

H
H

ALIF SEMICONDUCTOR

l c c R c A c

A l

-
-

A

-
-

A l

-

A l

-

-
- **NOTE**

A l

-

A

- H

A l

-

A rl A Al a al dfft rfflft c c l

-
-

A co c r c c l

-

H
H

A Al a l ac A a fft l

-
-

A fft A l r cr fftc l

-
-
-
-
-
-
-
  -
-

A DD c

-

l c Al r fft c

- H
-
-
-
-
-
-

l c l ffatfcfct

-
-

H
H

l c c fft c fft al l fft c

l c l fft

A l r c R rc rc a c

R rc r fft

```
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aisr=0x2
```

H

H

l c c fft fftr fft

-
-
-

A l a l fatcct

-

H
H

A rc r ft c c c

c a rl fftrffft l rffft

```
#define
#define
#define
```


Aft fft A a

-
-
- H

H
H

fft fft fft r ftc c

fft fft r ftc c d l A

-
-

fft fft r ftc c d l

-
-

H
H

ﬃt ﬃt ﬃt r ﬅc l c c A

ﬃt ﬃt ﬃt r ﬅc l c c R A

ﬃt ﬃt ﬃt r ﬅc l c c A

P

ﬃt ﬃt ﬃt r ﬅc l c c R A

H
H

P

H
H

r fft c c

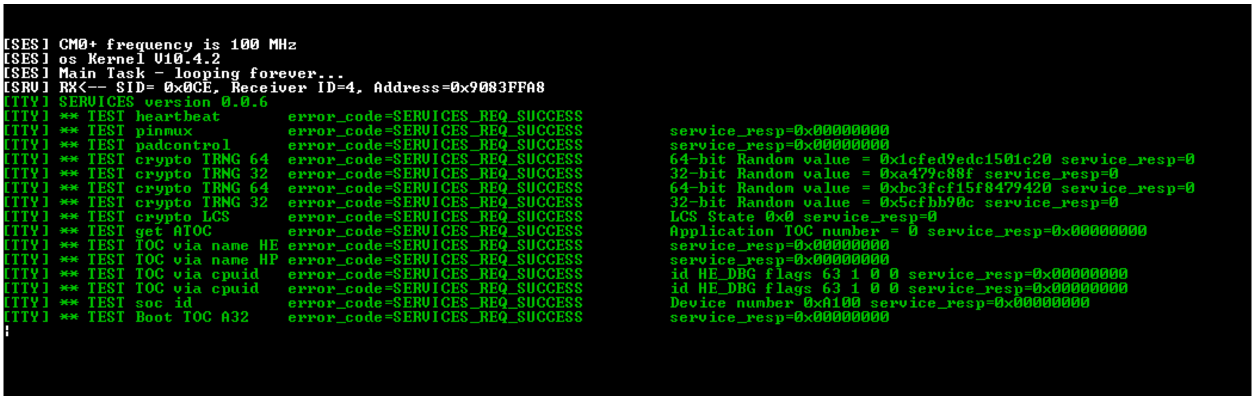
r fft c c rl fttc c r l a rfft

H
H

```
$ cmake .. -G "Unix Makefiles" -DENSEMBLE_CORE=M55_HE -DCMAKE_TOOLCHAIN_FILE=toolchain-armclang.cmake -DINSTALL_DIR=../junk
-- [INFO] Version=9
-- [INFO] installation override, using ../junk
-- The C compiler identification is ARMClang 6.18.2
-- The CXX compiler identification is ARMClang 6.18.2
-- The ASM compiler identification is ARMClang
```

H

H


[SES] CM0+ frequency is 100 MHz
[SES] os Kernel V10.4.2
[SES] Main Task - looping forever...
[SRV] RX<-- SID= 0x0CE, Receiver ID=4, Address=0x9083FFA8
[TTY] SERVICES version 0.0.6
[TTY] ** TEST heartbeat error_code=SERVICES_REQ_SUCCESS
[TTY] ** TEST pinmux error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTY] ** TEST padcontrol error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTY] ** TEST crypto TRNG 64 error_code=SERVICES_REQ_SUCCESS 64-bit Random value = 0x1cfed9edc1501c20 service_resp=0
[TTY] ** TEST crypto TRNG 32 error_code=SERVICES_REQ_SUCCESS 32-bit Random value = 0xa479c88f service_resp=0
[TTY] ** TEST crypto TRNG 64 error_code=SERVICES_REQ_SUCCESS 64-bit Random value = 0xbc3fcf15f8479420 service_resp=0
[TTY] ** TEST crypto TRNG 32 error_code=SERVICES_REQ_SUCCESS 32-bit Random value = 0x5cfbb90c service_resp=0
[TTY] ** TEST crypto LCS error_code=SERVICES_REQ_SUCCESS LCS State 0x0 service_resp=0
[TTY] ** TEST get ATOC error_code=SERVICES_REQ_SUCCESS Application TOC number = 0 service_resp=0x00000000
[TTY] ** TEST TOC via name HE error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTY] ** TEST TOC via name HP error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTY] ** TEST TOC via cpuid error_code=SERVICES_REQ_SUCCESS id HE_DBG flags 63 1 0 0 service_resp=0x00000000
[TTY] ** TEST TOC via cpuid error_code=SERVICES_REQ_SUCCESS id HE_DBG flags 63 1 0 0 service_resp=0x00000000
[TTY] ** TEST soc id error_code=SERVICES_REQ_SUCCESS Device number 0xA100 service_resp=0x00000000
[TTY] ** TEST Boot TOC A32 error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000

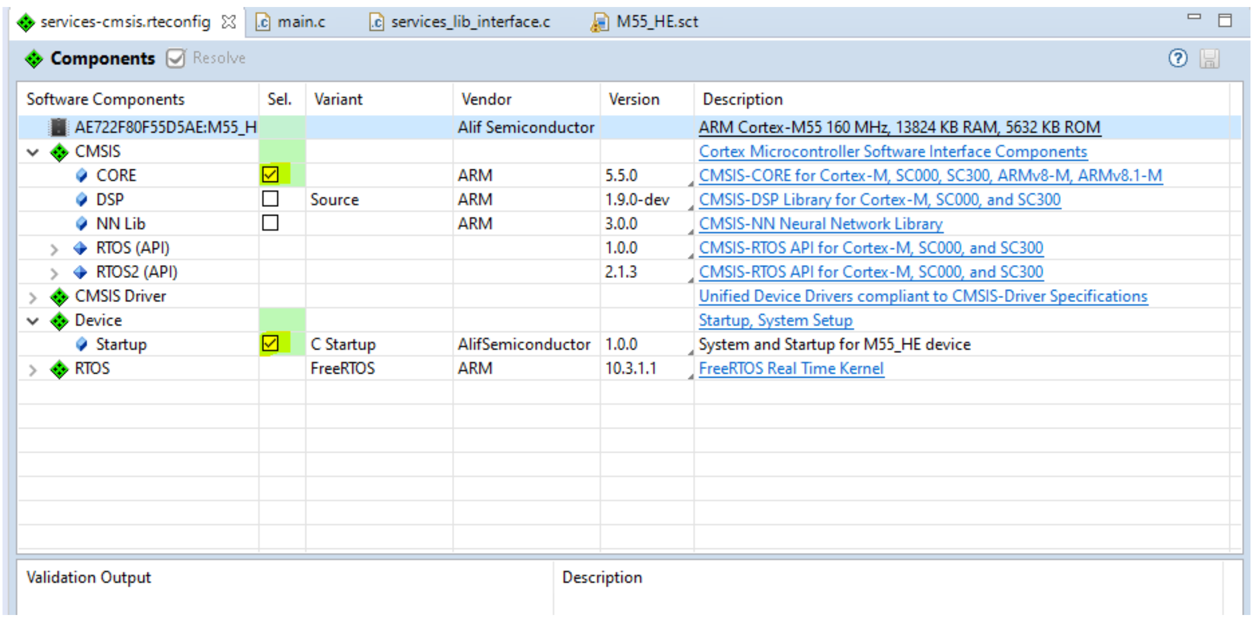
H

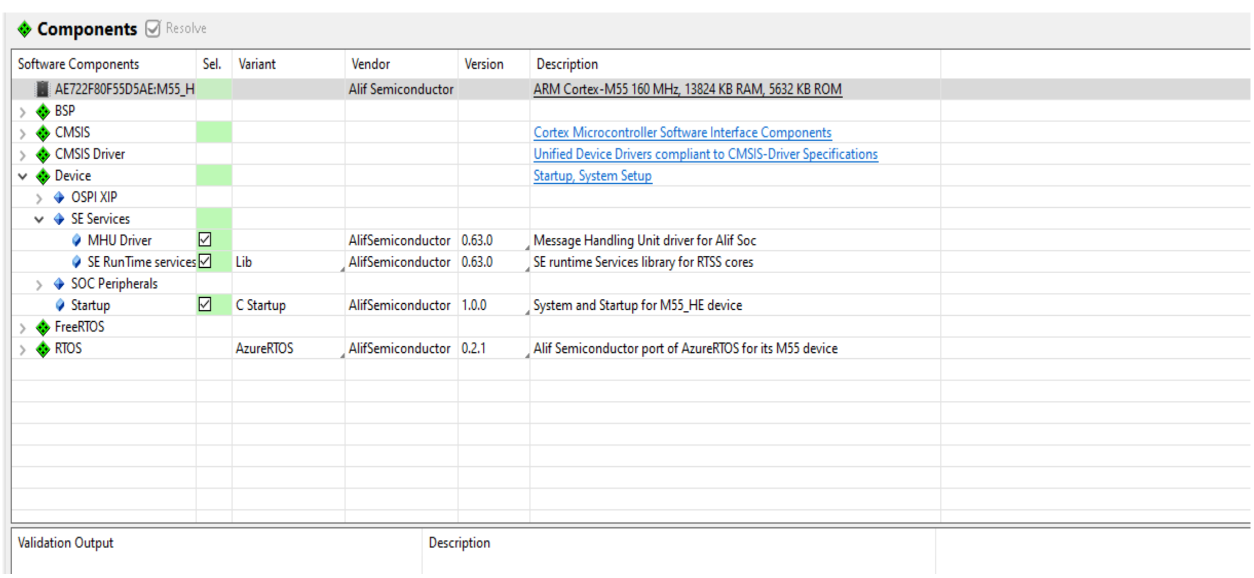
H

fftfft r ftc l r c c

H

-
-
-





H
H

fft D A rl l fat r fft al c

-
-
-

l r c fatc fft

-
-
-
-

H
H

l r c fffac fft rc d ac l rfft c

A fft

A c ft c c rl c

l r c fffac fft c

A fftfffftfftc

H

H

A c co c r

A c a c a a

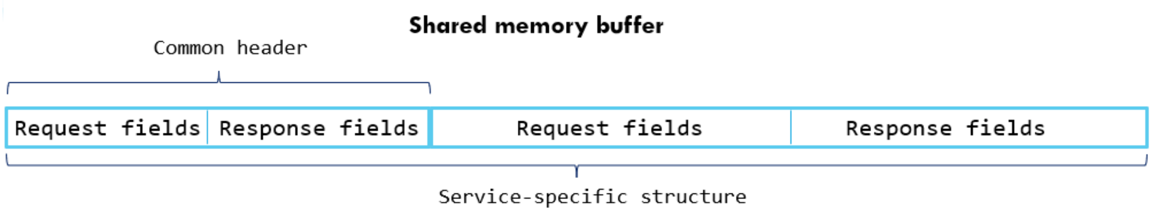
A a a

H

H

```c
typedef struct
{
  service_header_t header;
  volatile uint32_t send_<param>;      /*!< Send   parameter */
  volatile uint32_t resp_<param>;      /*!< Return parameter */
  volatile uint32_t resp_error_code;   /*!< Call error code  */
} example_service_t;

void SERVICES_example_call(services_req_t *service)
{
  example_service_t *p_svc =
    (example_service_t *)service->pkt_buffer_address; /* services request */
  uint32_t error_code;
  uint32_t local_result;

  error_code = function_call(p_svc->send_<param>, &local_result);

  p_svc->resp_<param>     = local_result;
  p_svc->resp_error_code = error_code;

  SERVICES_send_response_code(service, SERVICES_REQ_SUCCESS);
}
```

l r c fftc fft l r l rl al cr fft



Common header format

Service ID: req
Flags: req
Error Code: resp
Padding

```c
typedef struct
{
  uint16_t send_sid;
  uint16_t send_flags;
  uint16_t resp_error_code; // transport
layer error code
  uint16_t none_padding;
} service_header_t;
```

Service-specific structure example

```c
typedef struct
{
  service_header_t header;
  uint8_t send_port_num;
  uint8_t send_pin_num;
  uint8_t send_config_data;
  uint8_t resp_error_code; // service-specific
error code
} pinmux_svc_t;
```

l r c fftc fft l r l Al c

H
H

l r c fftc fft l fft

-
-

```
#define
#define
#define
#define
#define
#define
#define
```

l r c fftc fft c l ft fft

H

H

H

l r c ffac

ffac cl

A fffftfftc

- H
-

-
-
-

```
static ________
                                          __attribute__
int SERVICES_print const char

  return___
```

H

H

SERVICES wait ms

return

if

return

H

H

A c ﬃt

H
H

```
#define

#define
#define
```

H
H

A c c a cr dbc

H
H

A ftc r c r

```
if
  return

return
```

H

H

int

H
H

rc c c r

A rc cr c ffftc c

SERVICES_system_set_services_debug

H
H

ALIF SEMICONDUCTOR

A rc c l r

SERVICES_system_read_otp

if

return

H

H

A rc fftc l r

SERVICES_system_write_otp

A rc cr l r r

SERVICES_system_get_otp_data

H
H

if

return

A rc cr rl a r

SERVICES_system_get_toc_data

typedef struct

typedef struct

H
H

```
if

  return
```

```
[TTY] ** TEST TOC get data      error_code=SERVICES_REQ_SUCCESS          TOC number = 5 service_resp=0x00000000
[TTY] +--------------------------------------------------------------------------------+
[TTY] |    Name    |     CPU    |Load Address|Boot Address|Image Size|  Version   | Flags |
[TTY] +--------------------------------------------------------------------------------+
[TTY] |    DEVICE |  CM0+      | 0x00000000 | 0x00000000 |       296 |       0.5.0| u V   |
[TTY] |    DEVICE |  CM0+      | 0x00000000 | 0x00000000 |       372 |       0.5.0| u V   |
[TTY] |    SERAM0 |  CM0+      | 0x00000000 | 0x00000000 |     57612 |     1.105.0| u s   |
[TTY] |    SERAM1 |  CM0+      | 0x00000000 | 0x00000000 |     57612 |     1.105.0| u s   |
[TTY] | SRV-HE-T  |  M55_HE    | 0x58000000 | 0x58000000 |     73440 |       1.0.0| uLVB  |
[TTY] +--------------------------------------------------------------------------------+
```

H
H

SERVICES_system_get_toc_number

```
if
  return
```

H

H

SERVICES_system_get_toc_version

H
H

A rc cr rl a fft c

_________ SERVICES_system_get_toc_via_name _________

const ________

H
H

__________ SERVICES_system_get_toc_via_cpuid __________

```
typedef enum
```

```
  if
    return
     H
   H
```

H
H

SERVICES_system_get_device_part_number

0x0000B200

```
if
  return
```

H
H

SERVICES_system_get_device_data

typedef struct

H
H

```
if

  return
```

H
H

SERVICES_get_se_revision

H
H

A rc cr c fftc rc ffft

SERVICES_system_get_eui_extension

A rc cr c fatc fft

SERVICES_system_get_device_id64

H

H

```
if
  return
```

H

H

A fft

```
if
  return
```

A al r l

H
H

```
if
  return
```

A ffat r ffft l ffft fftc c

H

H

H
H

A l c rl l c co c r

```
  if
    return
```

H

H

```
if
  return
```

H
H

```
typedef enum
```

```
typedef enum
```

H
H

```
if

  return
```

H
H

```
#define
#define
#define

#define

#define

#define

#define

#define

#define
#define



typedef enum
```

H
H

if

return

H
H

```
typedef enum
```

H
H

```
if

  return
```

H
H

```
typedef enum
```

H

H

```
if

  return
```

H

H

H
H

```
if

  return
```

H
H

```
if
  return
```

H

H

A cr ad

```
if
  return
```

H
H

```
if
  return
```

H
H

```
if
  return
```

H
H

```
if
  return
```

H
H

A l c a a l r c al r l

A l c l l r c al r l

H
H

A l c crr fft al dfft c

```
typedef enum
```

H
H

H

H
H

Al a c ffatc

Al a dco c a cdfftfrffft

H
H

H
H
H
H
H
H
H
H

A a l a c car l a l ac

H
H
H

H
H

H
H

H
H
H
H
H
H
H

H
H

A al a cr dco c a

A al a cr dco c a

H
H

A al a c car l ac

H

H
H

H

H
H

H

H

H

A al a cr dco c a

H

H
H

H

H
H

H
H

A r r r

A r rl

H
H

A r fft r rc

A a r r

H

H

A a rl

A a fft l a c

A fftffffftfftc

H

H

A c fftuﬀt

l l r c ﬀatc

H
H

A llr lac rla c r

A llr a

H
H

H
H

A l l r cr rl

H

A l l r c cr a

H

H
H

A llr cc c a

H

H

A llr c cr l a

H

H

A rl c ffatc

A a rl ac cr

H
H

```
if
  return
```

A a rl ac cr a

```
if
  return
```

H
H

c c ffatc

SERVICES_cryptocell_mbedtls_hardware_poll

H

H
H

SERVICES_cryptocell_mbedtls_aes_init

SERVICES_cryptocell_mbedtls_aes_set_key

SERVICES_cryptocell_mbedtls_aes_crypt

H

H

SERVICES_cryptocell_mbedtls_sha_starts

H

H

SERVICES_cryptocell_mbedtls_sha_process

SERVICES_cryptocell_mbedtls_sha_update

SERVICES_cryptocell_mbedtls_sha_finish

H

H

SERVICES_cryptocell_mbedtls_ccm_gcm_set_key

SERVICES_cryptocell_mbedtls_ccm_gcm_crypt

H

SERVICES_cryptocell_mbedtls_chacha20_crypt

H

H

SERVICES_cryptocell_mbedtls_chachapoly_crypt

SERVICES_cryptocell_mbedtls_poly1305_crypt

SERVICES_cryptocell_mbedtls_cmac_init_setkey

H
H

SERVICES_cryptocell_mbedtls_cmac_update

SERVICES_cryptocell_mbedtls_cmac_finish

SERVICES_cryptocell_mbedtls_cmac_reset

H
H

R rc c fftc

A rc rl a

-
-
-
-

H

H		
H		

H

H

H
H

H
H

A ft r l cr l a

H
H
